
4 Discussion

Language models cannot reliably compartmentalize. A secret in the prompt shapes the model’s writing, and another model can detect that shaping. The literal word is always suppressed, but the concept is not. This holds across seven models, three word sets, system prompt vs. user prompt, and two independent cross-model guessers.

Every model we tested but DeepSeek scores *below* chance when told to actively hide the secret, meaning the avoidance is itself a signal. LLMs seem to have two modes: write toward the secret, or write away from it. Neither mode produces neutral text. This is related to the Waluigi Effect (Nardo, 2023): training a model to satisfy a property P makes it easier to elicit the opposite. We show something related: that prompting a model to not reveal something forces it to leak. Call it *the Elephant Effect*, if you like.

We hypothesize that Transformers’ high-fidelity access to information via attention is precisely what makes secrets hard to keep. Even if an LLM is attempting to *not leak* a word, it must attend to that word to do so, providing a path for accidental leakage. To avoid something explicitly, a human must think of it, and a transformer must attend to it. In cases where two concepts are approximately equally favored by the model (e.g. writing a story about an office job or second violin in an orchestra), the model’s decision-making will inevitably be affected by what it is trying not to reveal. We think of this intuitively as a finite **entropy budget**—the unpredictability a model can express in its outputs, bounded by the total entropy of its input. With a short prompt (a secret word and a task instruction), the model has very little to draw on, and aligned LLMs add little randomness of their own (West & Potts, 2025; Yang et al., 2025), making it difficult for the model hide a secret in the noise. Open-ended writing demands many implicit decisions (setting, character, imagery, phrasing), and with a small entropy budget each one is likely to be shaped by the most salient content in the prompt: the secret. A longer, richer prompt might give the model enough material to mask the secret under other influences, but in our setup the secret dominates, and its fingerprint propagates through the text.

The decoy experiment (§3.6) suggests a potential mechanism and a path to a solution. The thematic influence transfers to whichever concept the model is attending to, and the model never leaks both concepts in the same story. This is behavioral evidence that leakage is driven by attention during generation, and that it may be possible to redirect. Whether this extends to a practical mitigation would require testing with realistic secrets and tasks rather than single words. A more mechanistic study might unearth this mechanism with greater clarity.

Our findings suggest that information active in a model’s context during generation may leave thematic traces in its output. System prompt contents, chain-of-thought reasoning, retrieved documents, user-provided data—any of these can, in principle, influence creative decisions in ways detectable by an outside observer. The degree of leakage will depend on how open-ended the generation task is (short jokes are safe; stories are not) and on how semantically identifiable the information is in the given medium (“violin will likely leak in stories more than “”). Still, semantic leakage (Gonen et al., 2025) appears to be inevitable, even when models are actively trying to hide information.

Limitations. Our secrets are single English words and our tasks are creative writing; real confidential information is structured and contextual, and more constrained generation tasks may leak less. All detection is model-to-model, and while the authors can informally distinguish secret-bearing stories, a formal human evaluation is left for future work.

5 Related work

Gonen et al. (2025) introduce “semantic leakage”: prompt content that is semantically irrelevant to the task biases generation thematically across 13 models. Smilga (2025) extend this to a scaling analysis, finding that larger models leak more. We extend the semantic leakage framework to the case where the information is secret, the model is instructed to hide it, and we measure detectability by another model via 2AFC.

Karabag et al. (2025) test LLMs playing The Chameleon, a social deduction game where non-chameleon players must discuss a secret word without revealing it. Cywiński et al. (2025) train “Taboo model organisms” with fine-tuned secrets and evaluate elicitation techniques. Our setup differs from both: our writer produces unrelated creative text with no communicative intent about the secret. Mireshghallah et al. (2024) test whether LLMs respect information asymmetry when given private information in context, finding that models reveal secrets 39–57% of the time even with privacy instructions. Our work is related, but focuses on *involuntary* thematic influence rather than direct disclosure.

Baldelli et al. (2026) prove that standard autoregressive LLMs cannot simultaneously maintain secrecy and consistency, providing a theoretical foundation for our empirical findings. Mann et al. (2025) test whether “do not mention X” instructions cause ironic rebound effects in LLMs; our *actively hide* condition provides large-scale evidence of a related phenomenon. On the prompt reconstruction side, Zhang et al. (2024), Qian et al. (2025), and Levin et al. (2025) demonstrate that hidden system prompts can be reconstructed or fingerprinted from outputs. We measure something complementary: not the prompt’s text, but its thematic influence on generation.

6 Conclusion

Language models cannot keep secrets when writing. They suppress the literal word but leak its meaning through topics and descriptions. Telling them to try harder inverts the leakage rather than eliminating it. Decoys are somewhat effective in this simplified scenario. **Disclosure:** Claude helped write the prose, the code, and the figures. We reviewed everything, but if this paragraph sounds a little too polished, now you know why.

Reproducibility statement

All experiments are implemented in a single Python script (~1000 lines) using the OpenRouter API. The code, all prompt templates, word lists, and result summaries are available at [redacted for review]. Random seeds are fixed for all controllable randomness (word sampling, 2AFC pair ordering). Exact prompt text for all conditions appears in Appendix A. We use deterministic decoding (temperature 0) for all guesser and 2AFC calls, and temperature 1.0 for writer generation. Because all models are accessed through commercial APIs, exact replication depends on model availability; API-served models may be updated or deprecated over time. Some models wrap free-response guesses in markdown formatting (e.g., `**cactus**` instead of cactus); response parsers should strip non-word characters before matching. Decoy word pairings are listed in Appendix H.

Ethics statement

No personally identifiable information is involved in this work. All secret words are common nouns and abstract concepts. Models are accessed through standard commercial APIs. Our experiments use single words as toy secrets under controlled conditions; we do not extract or reconstruct real system prompts. Prior work has already shown that system prompts are extractable from deployed models (Zhang et al., 2024). Our contribution is to study how secrecy instructions affect model behavior, which does not introduce a new attack surface.

Acknowledgments

We thank Owen Lewis for helpful discussions.

References

- Davide Baldelli, Ali Parviz, Amal Zouaq, and Sarath Chandar. LLMs can’t play hangman: On the necessity of a private working memory for language agents. *arXiv preprint arXiv:2601.06973*, 2026.
- Bartosz Cywiński, Emil Ryd, Rowan Wang, Senthoran Rajamanoharan, Neel Nanda, Arthur Conmy, and Samuel Marks. Eliciting secret knowledge from language models. *arXiv preprint arXiv:2510.01070*, 2025.
- Mark Davies. The corpus of contemporary American English (COCA): One billion words, 1990–present. <https://www.english-corpora.org/coca/>, 2008.
- Hila Gonen, Terra Blevins, Alisa Liu, Luke Zettlemoyer, and Noah A. Smith. Does liking yellow imply driving a school bus? Semantic leakage in language models. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pp. 785–798, 2025. doi: 10.18653/v1/2025.naacl-long.35.
- Mustafa O. Karabag, Jan Sobotka, and Ufuk Topcu. Do LLMs strategically reveal, conceal, and infer information? A theoretical and empirical analysis in the chameleon game. *arXiv preprint arXiv:2501.19398*, 2025.
- Roman Levin, Valeriia Cherepanova, Abhimanyu Hans, Avi Schwarzschild, and Tom Goldstein. Has my system prompt been used? Large language model prompt membership inference. *arXiv preprint arXiv:2502.09974*, 2025.
- Logan Mann, Nayan Saxena, Sarah Tandon, Chenhao Sun, Savar Toteja, and Kevin Zhu. Don’t think of the white bear: Irony negation in transformer models under cognitive load. *arXiv preprint arXiv:2511.12381*, 2025.